Jingjing Pan

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POSITION

Humboldt Postdoc Fellow, **Karlsruhe Institute of Technology** 2025.10.1

Research Affiliate, **Lawrence Berkeley National Lab** 2022.5.1 - 2025.9.30

EDUCATION

Ph.D. in Physics 2018 Sep. - 2024 Dec.
Yale University, New Haven, CT

- Dissertation: *Exploring the Standard Model and Beyond through the Lens of Jet Substructure and Deep Learning with the ATLAS Detector*
- Advising Committee: Profs. Keith Baker, Helen Caines, Sarah Demers, Ian Moult, Benjamin Nachman

B.S. in Physics, Math 2015 Sep. - 2018 Aug.
Pennsylvania State University, State College, PA

- GPA: 3.95/4.00
- Advisor: Mikael Rechtsman

RESEARCH

The ATLAS Experiment at the LHC 2019-present

- **First measurement of jet track functions**, Analysis Contact
  - First application of an unbinned, high-dimensional unfolding method OmniFold for moment measurements
  - Developed the analysis to also first measure the renormalization group flow of jet track functions
  - **Responsible for:** sample production, statistical analysis, plotting framework, physics interpretation, analysis strategy, drafts editing
- **Machine learning applications**
  - Improved the precision, robustness, efficiency of OmniFold and deployed it in the full ATLAS Run2 dataset
  - Co-invented a normalizing flow-based method Neural Posterior Unfolding, which can give fast access to the full posterior via amortized learning
- **Hidden sector search via the Higgs portal**, Analysis Contact
  - **Responsible for:** signal MC validation, sample production framework, cutflow studies, plotting framework development

- **Jet calibration for the HL-LHC**
  - Performed pile-up mitigation and jet energy scale & resolution calibration, implemented for ATLAS Phase 2 Upgrade studies

The H1 Experiment at HERA 2023-present

- **First analysis of unfolding full phase space of H1 data**
  - First measurement of all particles in the H1 data
  - First measurement of Energy Correlators beyond the laboratory frame

Computational Photonics 2017-2019

- **Trapping light in an open system**
  - Predicted a photonic bound state enabled by the symmetries of an open system (that can be used for new efficient laser).

AWARDS/ FELLOWSHIPS

- Research affiliate with the ML4HEP group at LBNL 2021-present
- US ATLAS ATC Research Award (for visit at LBNL) 2022
- Lead Teaching Fellowship (Yale Physics) 2019
- Yale Early Start Research Grant 2018
- Downsbrough Chair in Physics Research Award 2018
- Department of Physics Teaching Award (1st undergrad recipient) 2018
- Bert Elsbach Scholarship (for academic performance) 2017
- Sigma Pi Sigma (Physics honor society) 2017
- Dean's List 2015-2018

PLENARY TALKS

Measurement of jet track functions with OmniFold-based binning corrections with the ATLAS experiment ML4Jets, Paris, 11/2024

Jet substructure measurements in multijet production with the ATLAS experiment BOOST, Genoa, 08/2024

Probing higher order DGLAP evolution using OmniFold and Energy Correlators BOOST, Berkeley, 08/2023

INVITED TALKS

Measurement of Jet Track Functions with OmniFold-based Binning Corrections in ATLAS Run 2 Data ATLAS ML Forum, 2/2025

Enabling Specialized Unfolding Methods with Modern Deep Learning Techniques France-Berkeley PHYSTAT Workshop, Paris, 06/2024

Improving and Extending Precision Jet Substructure Measurements with Deep Learning Techniques LBNL HEP Seminar, 6/2024

Improving and Extending Precision Jet Substructure Measurements with Deep Learning Techniques — Yale Wright Lab Seminar, 04/2022

Overview of Jet Calibration Techniques on ATLAS (session co-chair) — Hadronic Calibration Workshop, 08/2021

TEACHING

Yale Univeristy

- Mathematical Methods — Fall 2022
  - Guest lecture on Group Theory (notes)
- Nuclear and Particle Physics — Spring 2022
  - Guest lecture on research in ML techniques for jet studies
- Classical Mechanics — Fall 2021
  - Guest lecture on Path Integrals (notes)
- Introductory Physics Lab — Fall 2018-Spring 2020
  - Lead Teaching Fellow
  - Designed exam rubrics, organized grading, facilitated exam reviews and lab sessions, graded reports

Pennsylvania State University

- Statistical Mechanics — Spring 2018
  - Led recitations and reviews, helped with assignments
- Introductory Modern Physics — Spring 2018
  - Helped with assignments and held reviews
- Theoretical Mechanics — Fall 2017
  - Wrote psets and led in-class practice sessions

MENTORSHIP

Yale Univeristy

- Nathan Suri (graduate student, formerly at Caltech) — 05/2022-
  - *ML Pileup Mitigation for HL-LHC*
- Neal Ma (undergrad in Physics & Computer Science) — 04/2022-
  - *Optimizing step2 of OmniFold*
- Vinicius Da Silva (undergrad, now at Fermilab) — 10/2020-07/2021
  - *Search for $H \to Za \to 4l$ ($l = e, \mu$) in ATLAS Run2 data*
- Asmaa Aboulhorma (grad student in Morocco, on Z_d) 09/2021-05/2022
- Zainab Soumaimi (grad student in Morocco, on Z_d) 06/2021-01/2022

LANGUAGES/ SKILLS

Coding: Python, C++, Scheme (functional), MATLAB

Tools: GNU/Linux, ROOT, NumPy, Uproot/Awkward, pandas, TensorFlow, Keras, PyTorch (basic), Matplotlib, LaTeX, Delphes, Mathematica, Horovod, COMSOL, MPB, MEEP

Languages: English (proficient), Mandarin (native), Japanese (intermediate), Cantonese (intermediate)

OUTREACH/ SERVICE

- Reviewer, Machine Learning for Physical Sciences (ML4PS) Workshop (Neural Information Processing Systems (NeurIPS) Conference, 2024)
- Organizer, ATLAS Group Journal Club (Yale, 2022-2024)
- Moderator, Kimball Smith Series (Yale Jackson School of Global Affairs, 2023)
- TA, US ATLAS Machine Learning Training (LBNL, 2023)
- Mentor, Yale Women in Physics (2019-22)
- Reviewer, Proceedings A
- Organizer, Conference for Undergraduate Women in Physics (Yale, 2020)
- Attendant, Conference for Undergraduate Women in Physics (Princeton, 2017; Columbia, 2018)
- Volunteer, Science Booth Outreach at central PA Arts Fest (2017)